ORIENTATION TO SOFTWARE ENGINEERING / Final EXAM
(Slightly shortened DEMO EXAM. Use your imagination for rest of the tasks)

Given name:_____ Family name:_____ Number: _____

Part 1 alone gives you an *exam grade* max 3.

- But reaching 3 most likely also needs several tasks from part 2 programmed successfully.
- Each correct answer gives you one point. No answer = 0 points.

Part 1 and **Part 2** together can give you a higher *exam grade* (4 or 5).

DO NOT TURN PAPER BEFORE GET PERMISSION, THANKS!

- **Switch the phone off and place it in the bag, away from you.**
- **Put everything else than pencils, erasers, and water/beverages away.**
- **Put jackets, caps etc. far away.**
- **Place bags etc. 2 meters or more away.**
- **Keep eyes on your own paper, thanks.**
- **You may not take any photos, not even after the exam.**
- **Return all the papers.**

(Cheating or helping somebody in the exam or sharing individual homework would mean not coming back to this course before next semester, taking all the course fully again. No RPL or like that offered.)

You can shorten…

console.log("Hi!");

…as…

c.l("Hi!");

This of course doesn't work in any browser or so, just use it for our paper exams.

All in all if you feel that you'll need to write a lot in some task, not fitting even in the answer box, there most likely is simpler solution to the task.

MULTIPLE-CHOICE TASKS

1. Select ***one*** from the list below. Answer: _____

   At home, the thing that reminds an ***algorithm*** the most would be:
   a. electronic washing machine
   b. housewarming party
   c. remote control of TV or radio
   d. computer game
   e. programmable microwave oven
   f. food recipe with ingredients and steps
   g. programmable electronic calculator
   h. none of the above

2. Select ***one*** from the list below. Answer: _____

   In ***computer programming***, …..
   …

Plus 3-6 more questions like this. Some are technical, like variables, scopes, functions. Can be about principles, best practices in programming. Or about terms like variable, reference, and object. Or about algorithms and programming logic.

Also remember the Development life cycle and Process model chapters 11 and 12: SDLC, waterfall, agile, Scrum, Kanban, Software development process roles, … Usually 1-3 questions from there.

3. Select ***one*** from the list below. Answer: _____

   Abc
   …

4. Select ***one*** from the list below. Answer: _____

   Def
   …

5. Select ***all that apply***. Answers: _____

   Ghi
   …

6. Select ***all that apply***. Answers: _____

   Jkl
   …

CODE READING TASKS (Do not guess what happens => <u>Track</u> the variables and values <u>on paper</u>!)

7. What does the code below print to the console window? Answer: _____

```
var x = 1 + 2 * 3;
var y = x;
x = 8;
console.log(x + y);
```

8. What does the code below print to the console window? Answer: _____

```
var x = 499;
var y = 501;
x = y;
y = 500;
console.log(x);
```

9. What does the code below print to the console window? Answer: _____

...

10. What does the code below print to the console window? Answer: _____

…

11. What does the code below print to the console window? Answer: _____

```
var x = 15;
while (x >= 5) {
    x = x - 3;
}
console.log(x);
```

12. What does the code below print to the console window? Answer: _____

```
...
```

13. What does the code below print to the console window? Answer: _____

```
var row = [8, 3, 4, 5, 14, 12, 7, 9];
var col = [3, 2, 1, 4, 5, 7];
var x = row[col[3]];
console.log(x);
```

14. In JavaScript, the *modulo operator* is **%.** The result of the modulo operation is the *remainder of the division*. examples:

    10 % 6 is 4
    12 % 3 is 0
    10 % 100 is 10

    What does the code below print to the console window? Answer: _____

```
for (var fill = 6; fill < 9; fill++) {
    console.log(fill % 7);
}
```

15. What does the code below print to the console window? Answer: _____

```
var a = {x: 5, y: 3, z: 1};        // Here we have a JavaScript object
var b = {x: 2, y: 9, z: 1};        // And here another

b.x = 7;
b = a;
a.x = a.x + b.x;

console.log(b.x);
```

(POSSIBLE) DRAWING TASKS

16. How to draw the variables, variable values, objects, and object references of these three separate codes?

(These were class examples, mostly from the course material code. Try to draw first and find the solutions drawn in Teams)

```
var country = {name: "Finland", population: 5501043};
```

```
var myCountry = {
 name: "Finland",
 population: 5501043,
 capital: {name: "Helsinki", population: 634509}
};
```

```
var countries = [
 {name: "Finland", population: 5501043},
 {name: "Sweden", population: 9747355},
 {name: "Norway", population: 5165802}
];
```

(POSSIBLE) PROGRAM PART IDENTIFICATION TASKS

17. Task type one: Connect the identifiers 1-4 to the terms a-e (one term will not be used)

The identifiers are confusing on purpose. Programmers have to be able to identify parts of program no matter what they are called. Aa, bee, cee, dee. Or helsinki, stockholm, oslo, tallinn.

You identify the parts based on where they are put and how.

```
foo = bar.sna(fu);
```

terms
a. parameter
b. object
c. variable
d. scope
e. function

identifier	**Selected term a-e:**
1. foo	Answer:_____
2. bar	Answer:_____
3. sna	Answer:_____
4. fu	Answer:_____

18. Task type two: Connect the functions 1-4 to the function types a-d (all will be used exactly once)

a. function that is given nothing and that returns nothing
b. function that is given nothing and that returns something
c. function that is given something and that returns nothing
d. function that is given something and that returns something

1. var element = document.getElementById(“txtName”)	Answer:_____
2. console.log(“Hello “ + name + “!”)	Answer:_____
3. number = Math.random()	Answer:_____
4. element.clear()	Answer:_____

(POSSIBLE) ERROR TRACKING TASK

19. (You are given a 10-15 line long very basic HTML file and a 5-10 line long basic JavaScript file that are supposed to work together. Your task is to identify and briefly list all errors you can find, in either file.

    Most of the errors are related to how the HTML document refers to the JavaScript code, and how the JavaScript code refers to things in the HTML document.)

No sample HTML or JavaScript given to you in this demo exam. But all in all, look at Function Examples, example 2. That is your best source for finding out what is the correct way to refer from HTML to JavaScript and from JavaScript to HTML file. Really crucial for many ways to learn that example well!

PROGRAMMING TASKS OF PART 1 – (Usually simpler, only testing one or two features)

20. In Evenlandia, everybody earns 10 euros per hour. Though after 8 hours the additional hours are considered overtime, with 15 euros per hour for those extra hours.

    Write an algorithm that finds out how much is the total salary. The hour input is already read for you. You can print the earned euros **to the console** just as one number. (Test case: 10.5h => 117,5 (€))

    - To simplify this task, you can assume you get valid text as input.

```
var hourInput = document.getElementById("txtHoursWorked").value;
```

21. Write a piece of code that finds the ***…sum/average/max/min/ (or something similar task)..*** *of the number values in the array below*. (Naturally, your code in all of our tasks must work with other values, too. Those below are just examples.)

```
var a = [1, -2, 3, 3, 4, -5, 2, -7, 3, 5, 3, 5, 4, 1, -3, 3, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, -1];
```

23. Define a function called **checkColor** which is given a name of a color as parameter. When called, the function will check if the color is one of these: “red”, “white”, or ”blue”. If the color is one of those, the function will return (Boolean value) *true*, otherwise the function will return the value *false*.

INSTRUCTIONS (to be followed carefully - otherwise, no points given)
- To simplify this task, you can assume that the function is given valid text as input
- The function should be totally independent of any HTML page.

24. **Call the** checkColor **function** (described above) under the already written code line below. Use the already read value below as input.

Also print the result, true or false, to console

INSTRUCTIONS (to be followed carefully - otherwise, no points given)
- To simplify this task, you can assume that you get valid text as input

```
var colorInput = document.getElementById("txtColorName").value;
```

PART 2 - *This part is needed **for sure** if aim at grades 4 or 5. And also for 3 if need to cover any lacks in the previous tasks. But relax and just do as many tasks as you can. All points are not required, not even for 5.*

There are two arrays, **players** (string array, names of the players) and **scores** (array of number values, the number points the user has gathered. The more the merrier.)

They are synchronized, so that using same index you get a player and her/his score. You can assume that there is at least one item in both arrays, and naturally same number of items in both. It is possible, that there are over thousand items in the arrays. The following questions use those arrays. Each task is independent on its own. You can do any of them or all of them. First ones are usually the easiest. But you can skip if stuck and try latter ones.

INSTRUCTIONS FOR REST OF THE TASKS (to be followed carefully - otherwise, no points given)

- To simplify this task, you can assume that the array(s) are valid, and the input values are *valid.*
- In this task, you are ***NOT*** allowed to call any already existing function or method. (Other than console.log, myArray.push/pop, naturally)
- Follow the requirements and do not do or print anything extra. **Extra code = points reduced!**
- Print to console with console.log( ) if instructions say just "to print" something.

25. Print the total number of points over all the players.

26. Complete the code so that you try to find the given player. If found, print the score for that player. If not found, print “Joe, Not found!” if ‘Joe’ was searched for.

```
var name = document.getElementById("txtPlayerName").value;
```

27. Print all the player names who have score higher than 10. If none, print nothing.

28. Print name(s) and score(s) of the players(s) that have the maximum score. Note that several might have that maximum.

29. Print name(s) and score(s) of the players(s) that have score higher than the player with the received name. If the name not found in the array then do nothing.

```
var name = document.getElementById("txtPlayerName").value;
```

30. Count the total points of the players starting from player with name name1 until player with name name2, including those players. If even one of them (name1 or name2) is not found, do nothing.

```
var name1 = document.getElementById("txtPlayerName1").value;
var name2 = document.getElementById("txtPlayerName2").value;
```

31. Define a function called **findIndexOfName** that is given an array of text strings (actually a reference to that array) and a name to find. The function will return the index of the searched name if in the array, or null if not found.
The function itself should not print anything. Test the function using "Kari" and the names array.

32. Do this task utilizing the **findIndexOfName** from previous task (imagine that it works if you did not get it working)
    Using an array of names of scorers, var scorers, go through it and mark one additional point to the original scores array. If name appears many times, then add multiple points.

33. There might be also **something students with grade 5 skills should try to do. Not actually difficult for anybody. Just different from many tasks you have seen.**
(Some task that is related to those cases, where the order of the item matters. E.g. days, Monday, Tuesday, Wednesday etc. Then teachers can create tasks with different logic, but still with same kind of array and loop(s) skills) – Not revealed here, think of all possibilities.

HOW WILL THE REAL EXAM DIFFER?

- probably no multi-choice in beginning

- a bit more of the code reading tasks

- more coding tasks, most likely 2-5 more of those two-three array programming logic tasks, like the players and points

- all tasks will be different. None will be from this demo exam.

- It will not e.g. be players and points, but something similar

- but, being able to solve this demo exam measures the same skills as the final exam